Microsoft Elevate

CAPSTONE PROJECT

IPL DATA ANALYSIS

Using Power BI

PRESENTED BY

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OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result (Output Image)
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT:

- IPL produces huge amounts of match and player data each season.
- Analyzing this data manually is time-consuming.
- Teams, analysts, and fans need interactive insights on matches, players, and trends

OBJECTIVE:

- Build an interactive Power BI dashboard to visualize IPL data and provide actionable insights..

PROPOSED SOLUTION:

Solution Approach:

- Use Power BI Desktop to connect, clean, and model IPL data.
- Create KPI cards, charts, and slicers for quick analysis.
- Provide interactive insights on matches, top scorers, toss decisions, and team performance.

SYSTEM APPROACH:

Workflow Diagram (Optional Graphic)

1. **Data Collection**: Download matches.csv & deliveries.csv
2. **Data Cleaning:** Remove empty rows, fix datatypes
3. **Data Modelling:** Create relationships (matches.id → deliveries.match_id)
4. **Visualization**: KPI cards, charts, pie chart, column chart
5. **Interactivity:** Add slicers for season, team, and venue
6. **Insights:** Derive meaningful trends

ALGORITHM :

1. Load CSV files into Power BI

2. Transform and clean the data (remove nulls, fix season column)

3. Create relationships between tables

4. Build KPI cards for total matches, runs, and sixes

5. Create charts:

- Matches won by team (**Bar Chart**)
- Top run scorers (**Bar Chart**)
- Runs per season (**Column Chart**)
- Toss decision analysis (**Pie Chart**)

6. Add slicers for interactive filtering

7. Add insights text box for observations

DEPLOYMENT:

- **Power BI Desktop File (PBIX):** Submit as offline project
- **Optional Online Deployment:** Power BI Service → upload PBIX → share link
- **Export PDF:** For static view

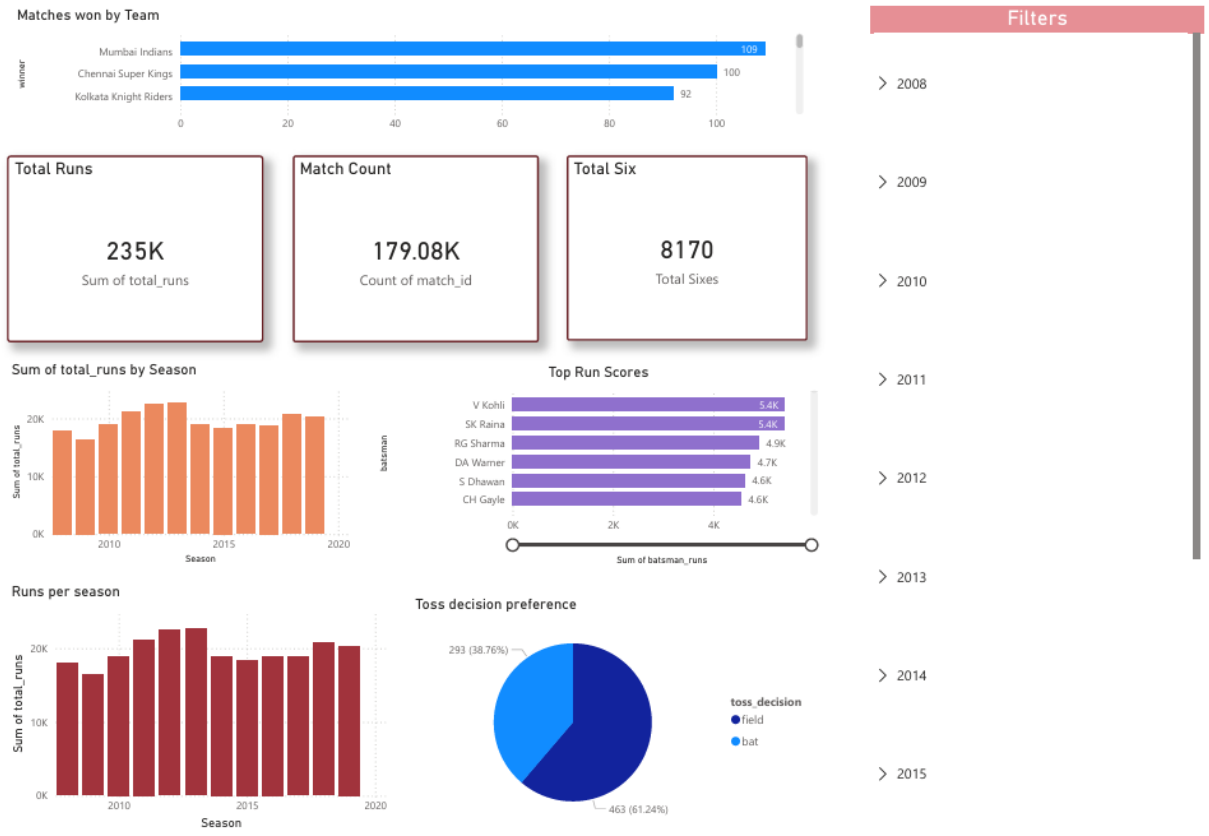
RESULT:

KPI Cards (Total Matches, Runs, Sixes)

Charts (Matches Won, Top Scorers, Runs per Season, Toss Decision)

Slicers (Season, Team, Venue)

Insights section



- **Mumbai Indians** and **Chennai Super** Kings have the highest wins.
- Most teams prefer **fielding** after winning the toss.
- Total **runs** have increased over recent IPL seasons.
- **Virat Kohli** and other top batsmen dominate scoring.

CONCLUSION:

- Successfully built an interactive IPL dashboard
- Can analyze matches, teams, top players, and trends easily
- Interactive slicers allow filtering by season, team, or venue
- Provides quick insights for analysts, fans, and teams

FUTURE SCOPE:

- Add player vs player comparison charts
- Predict future match outcomes using Machine Learning
- Include venue-wise performance analysis
- Integrate with live IPL data APIs for real-time dashboard

REFERENCES:

GitHub Link: Github Link

CVS Link: CSV Files Link

PBIX File Link: PowerBi File link

Result PDF Link: Output PDF

LinkedIn Link: Linkedin Profile

Thank You